

Do Vision-Language Models Bind Spatial Relations?

A Train-Free Diagnostic with No-Supplied-Box Spatial Prompts

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Abstract

Spatial questions can be answered either by binding the queried relation in the image or by exploiting object and language priors, and aggregate VQA accuracy often hides which behavior occurred. We evaluate this ambiguity with a train-free paired diagnostic over 4,460 scored responses from a gpt-5-mini hosted multimodal backend on What’sUp, SRBench, and SpatialMQA, using the same benchmark harness, parser, exact-match metrics, and 512-token response budget for every condition. For each selected task, the harness runs direct visual answering, natural-language chain-of-thought, constraint-aware prompting, a coordinate/box-oriented prompt that receives no supplied boxes, and a no-image prior diagnostic, so the comparison tests prompt-conditioned behavior rather than weight adaptation or localization supervision. The no-supplied-box coordinate/box-oriented prompt scores 76.91% pooled accuracy, essentially matching direct visual answering at 77.02% and trailing constraint-aware prompting at 78.70%; its paired delta against direct visual answering is -0.11 percentage points with a 95% bootstrap interval of [-1.79, 1.57]. The contribution is therefore an evidence-bounded diagnostic that separates a measured null result from stronger supplied-box grounding, flip-consistency, ablation, and robustness claims that require new box-bearing or pair-audited evidence.

Introduction

Vision-language models (VLMs) often appear capable on image-question tasks while still struggling with spatial relations such as left of, right of, above, below, in front of, behind, and containment. Controlled spatial-relation benchmarks such as What’sUp isolate this behavior by holding object identities fixed while varying relation labels (Kamath, Hessel, and Chang 2023). Compositional probes such as Winoground and VSR show that visually grounded systems can fail when language and image structure must be aligned rather than recognized independently (Thrush et al. 2022; Liu, Emerson, and Collier 2022). Spatially annotated resources and recent MLLM-specific benchmarks broaden the same concern to relation-heavy visual reasoning settings (Stogiannidis, McDonagh, and Tsaftaris 2025; Liu et al. 2025; Jia et al. 2025). These resources make the empirical question concrete: a model can answer many spatial questions correctly while still leaving unclear whether it used image-grounded relation evidence or a prior over typical object arrangements.

This paper treats that ambiguity as a measurement problem rather than a training problem. We do not train or adapt model weights. Instead, we freeze a paired task matrix over three selected public benchmark families and evaluate a gpt-5-mini hosted multimodal backend under five inference-time conditions with the same parser and benchmark-specific exact-match scorer. The conditions are direct visual answering, natural-language chain-of-thought, constraint-aware prompting, a coordinate/box-oriented prompt, and a no-image prior diagnostic.

The coordinate/box-oriented condition is the key scope boundary. Such a prompt could test supplied-object grounding only if the benchmark release provides boxes or coordinates. The selected What’sUp, SRBench, and SpatialMQA task files provide zero supplied box rows and zero coordinate rows, and no detector boxes are introduced. We therefore evaluate it as a no-supplied-box prompt-shape diagnostic rather than as a localization-grounded method.

Across 892 paired task keys and 4,460 method responses, the no-supplied-box coordinate/box-oriented prompt reaches 76.91% pooled accuracy, compared with 77.02% for direct visual answering and 78.70% for constraint-aware prompting. Its paired delta against direct visual answering is -0.11 percentage points with a 95% bootstrap interval of [-1.79, 1.57]. The contribution is a calibrated negative/null result: it rules out the stronger supplied-box claim for this evidence pack, records where image access still matters relative to a no-image prior, and specifies the box-bearing, pair-audited, and ablation evidence required for stronger grounding claims.

Related Work

Spatial and compositional VLM evaluation. What’sUp isolates spatial relations by keeping object identities fixed while varying relation labels (Kamath, Hessel, and Chang 2023). SpatialMQA and SRBench probe MLLMs on broader spatial-relation question sets, while OmniSpatial and related benchmark-driven studies broaden the test space (Liu et al. 2025; Stogiannidis, McDonagh, and Tsaftaris 2025; Jia et al. 2025; Yu et al. 2025). Older visual reasoning resources such as CLEVR and Visual Genome remain important because they separate object, attribute, and relation structure (Johnson et al. 2016; Krishna et al. 2016). Recent failure analyses also caution that apparent image-language competence can coexist with brittle relation judgments (Wang et al. 2024). These

benchmarks motivate relation-specific evaluation rather than treating spatial accuracy as a side effect of generic VQA skill.

The diagnostic stance also follows a broader lesson from VQA and multimodal evaluation: dataset-level accuracy can improve without proving that the image supplied the decisive evidence (Antol et al. 2015; Goyal et al. 2017; Hudson and Manning 2019). General-purpose MLLM benchmarks expose broad capability profiles, but their aggregate scores still need task-specific audits when the claim concerns a narrow behavior such as spatial relation binding (Fu et al. 2023; Liu et al. 2023b; Li et al. 2023a).

Compositional image-text tests make the same caution more concrete. Winoground and VSR probe relation-sensitive alignment (Thrush et al. 2022; Liu, Emerson, and Collier 2022), while VL-CheckList, ARO, CREPE, and SugarCrepe stress object, attribute, and relation composition in image-text matching (Zhao et al. 2022; Yuksekgonul et al. 2023; Ma et al. 2022; Hsieh et al. 2023). The present paper narrows the question to spatial VQA-style tasks and keeps each method paired on the same task keys, so a prompt condition can be compared without changing the benchmark distribution.

Grounding, boxes, and spatial prompting. Object-level evidence can matter for grounded spatial reasoning (Rajabi and Kosecka 2023; Cheng et al. 2024). Region-caption, referring-expression, and attention-based VQA work show why object locations and phrase grounding are different evidence channels from ordinary image-level answering (Plummer et al. 2015; Kazemzadeh et al. 2014; Mao et al. 2015; Yu et al. 2016; Anderson et al. 2018; Johnson, Karpathy, and Fei-Fei 2016). Training-oriented work such as SpatialVLM and Sparkle studies how data or supervision can improve spatial capabilities (Chen et al. 2024; Tang et al. 2024). This paper is complementary: it asks what can be measured with train-free inference-time prompts and explicitly refuses to count a coordinate/box result as supplied-box evidence when no supplied boxes exist.

This distinction is not cosmetic. A supplied-box method can test whether explicit object locations help relation decisions; a no-supplied-box prompt can only test whether asking the model to reason in that format changes its answer distribution. The current benchmark bundle supports the latter experiment. It does not support claims about detector quality, center arithmetic, or object localization as a causal bottleneck.

Reasoning interventions. Modern vision-language systems build on large-scale image-language pretraining and instruction tuning (Radford et al. 2021; Li et al. 2022, 2023b; Liu et al. 2023a), so a train-free paper should separate benchmark behavior from model training claims. Constraint-aware prompting and chain-of-thought-style prompting are natural baselines for spatial tasks (Wu, Liu, and He 2025; Kancheti et al. 2026; Andreas et al. 2016). Our evidence treats them as comparable inference-time conditions, not as explanations by themselves. A prompt can change behavior without proving that the model localized objects or bound relations.

This is why the no-image prior is included as a diagnostic rather than as a competing visual system. It estimates how

often question text alone is enough under the same parser and answer vocabulary. The large visual-versus-no-image gap in the results shows that images matter for this selected matrix, while the absence of audited prior-conflict groups prevents a stronger claim about when priors override visual evidence.

Diagnostic Setup

Each selected task has a benchmark family, a question, a gold answer under that benchmark’s scoring rule, and a relation-family or source-stratum tag when available. The selected families are What’sUp controlled images, SRBench, and SpatialMQA. The evaluation contains 4,460 scored responses: 892 selected task keys crossed with five methods. What’sUp contributes 412 task keys, while SRBench and SpatialMQA contribute 240 each.

The five conditions are paired at the task level. Direct VLM answering receives the image and question. Natural-language chain-of-thought asks for a brief visible-relation rationale before the final answer. Constraint-aware prompting asks the model to check inverse and mutually exclusive relation constraints. The coordinate/box-oriented condition asks for object-location reasoning but receives no supplied boxes or coordinates because the selected task releases contain none. The no-image prior diagnostic removes image access and estimates text-only answerability for the same question form.

All visual conditions use the same gpt-5-mini hosted multimodal backend through a Responses-style interface. Each response is constrained to a JSON final answer and confidence field, with a maximum of 512 output tokens and low reasoning effort; temperature and top-p use the backend defaults. The benchmark harness builds one episode by loading the selected image and question when the condition allows image access, applying the condition-specific prompt template, parsing the returned final answer into the benchmark answer vocabulary, and assigning an exact-match score. The evaluation is train-free and inference-only: the method changes prompts and available image access, not model weights.

The evaluated system is therefore a five-condition diagnostic harness, not a trained model variant. Its fixed components are the selected public task sources, the gpt-5-mini hosted multimodal backend for all visual and no-image calls, the condition-specific prompt templates, a JSON final-answer parser, benchmark-specific exact-match scorers, and the 512-token low-effort response budget. The baselines are direct visual answering, natural-language CoT, constraint-aware prompting, and no-image answering; the tested intervention is the no-supplied-box coordinate/box-oriented prompt.

Benchmark Provenance and Scope

The benchmark bundle was selected to cover complementary spatial-relation regimes rather than to maximize a single aggregate score. What’sUp contributes 412 controlled-image tasks from the official controlled-images release, preserving its object-identity control and first-caption scoring policy. SRBench contributes 240 test-split examples sampled evenly across the selected source strata: figure-position, mental-rotation easy, mental-rotation hard, navigation, orientation, and relations. SpatialMQA contributes 240 test-split exam-

ples across six relation groups: behind, below, in front of, left of, on/above, and right of. The resulting 892 task keys are paired across all five conditions, so the main table compares prompts on identical task instances rather than on family-specific or method-specific subsets.

This provenance matters because the three sources expose different difficulty and auditability tradeoffs. What’sUp is controlled and relation-focused, but the selected visual backend is already near ceiling on the sampled controlled images. SRBench is broader and includes navigation, orientation, relation, and mental-rotation strata, so it is useful for detecting whether a method’s effect is specific to easy two-object relations. SpatialMQA gives a public MLLM spatial-relation test split with explicit relation labels, which supports relation-slice reporting but not full counterfactual consistency claims. The evaluation therefore treats the three families as a deliberately heterogeneous diagnostic matrix. A pooled result is reported only after each method has a row for each task key, and every family row remains visible so a reader can see whether one source carries the conclusion.

The box-source check is equally central. The selected task records contain zero supplied box metadata and zero coordinate metadata across all three families. No detector boxes, VLM-readout boxes, or post-hoc object centers are introduced. The coordinate/box-oriented condition therefore uses the same image and question as the other visual conditions but changes the requested reasoning format: it asks the model to reason about object locations while explicitly receiving no external geometry channel. This is why the paper uses the phrase no-supplied-box throughout the abstract, method, results, and conclusion.

The selected evidence can support three kinds of claims. First, it supports paired prompt-condition comparisons across 4,460 scored responses because every selected task is evaluated under the same five conditions. Second, it supports image-access comparisons against the no-image prior because the question text and answer parser remain fixed while image access is removed. Third, it supports descriptive relation and source-stratum analysis because relation-family tags and SRBench strata are recorded in the selected rows. The evidence does not support supplied-box grounding, detector quality, center-arithmetic mechanisms, flip-pair consistency, or robustness across backend versions. Those stronger claims would require a different package: box-bearing sources or detector outputs, audited counterfactual pair identifiers, component ablations, and repeated route or model-version runs.

This scope boundary is not a post-hoc defense of a negative result. It is the experimental design constraint that makes the negative result interpretable. If the no-supplied-box coordinate prompt had beaten direct visual answering, the correct interpretation would still have been prompt-conditioned behavior rather than verified localization. Since it does not beat direct visual answering or constraint-aware prompting, the conclusion is narrower and cleaner: in this public, paired, no-box matrix, a coordinate/box-oriented prompt shape is insufficient to improve spatial-relation exact match over ordinary visual prompting. The next experiment should change the evidence channel, not merely rename the prompt.

Methods

Figure 1 shows the method matrix. The design goal is paired comparison rather than method invention. All five conditions are evaluated on the same task keys, with the same answer parser and score semantics. Direct VLM is the default visual QA baseline; natural-language CoT and constraint-aware prompting are inference-time reasoning baselines; the no-supplied-box coordinate/box prompt is the tested location-oriented prompt; and the no-image prior is a text-only diagnostic rather than a visual method. Unrecoverable parse failures are scored incorrect rather than converted into plausible answers.

The harness normalizes each response to the benchmark answer vocabulary before scoring. For What’sUp, the metric is the official first-caption exact match over the controlled-image selection. For SRBench and SpatialMQA, the metric is the corresponding benchmark exact-match score over the selected test split. These metrics are then pooled only after every task key has one scored response for each condition.

Experimental Protocol

The primary metric is the benchmark-specific exact-match score normalized to task-level accuracy. Statistical comparisons use paired task keys. We report bootstrap confidence intervals over paired tasks and exact McNemar tests on discordant paired outcomes, with Holm correction across the comparison table. Parse-failure diagnostics are reported separately, but official accuracies retain those rows as incorrect.

The headline claim is evidence-limited by the available task metadata. All 4,460 expected responses were scored, all method-family groups were complete, and zero selected tasks included supplied box or coordinate metadata. Therefore all supplied-box, detected-box, center-arithmetic, and localization-bottleneck interpretations are outside this run.

The response budget is fixed across methods: at most 512 output tokens, low reasoning effort, and backend-default sampling controls. This keeps comparisons paired at the prompt-condition level while avoiding claims about sampling robustness or backend-version robustness, which are not evaluated here.

Validity Checks

The evaluation includes several validity checks that constrain how the results can be interpreted. First, the matrix is complete at the task-method level. Each of the 892 selected task keys contributes one scored response for each of the five conditions, yielding 4,460 scored rows. This prevents the main comparison from depending on uneven task coverage across methods. The same completion check also makes the pooled table meaningful: pooled accuracy is not estimated from different subsets for different prompt conditions.

Second, the comparisons are paired by task key. The direct VLM, natural-language CoT, constraint-aware, coordinate/box-oriented, and no-image rows for a task are evaluated under the same benchmark family and answer vocabulary. Paired bootstrap intervals and McNemar tests therefore ask whether two conditions differ on the same selected examples rather than whether two independently sampled

Train-Free Spatial Relation Diagnostic

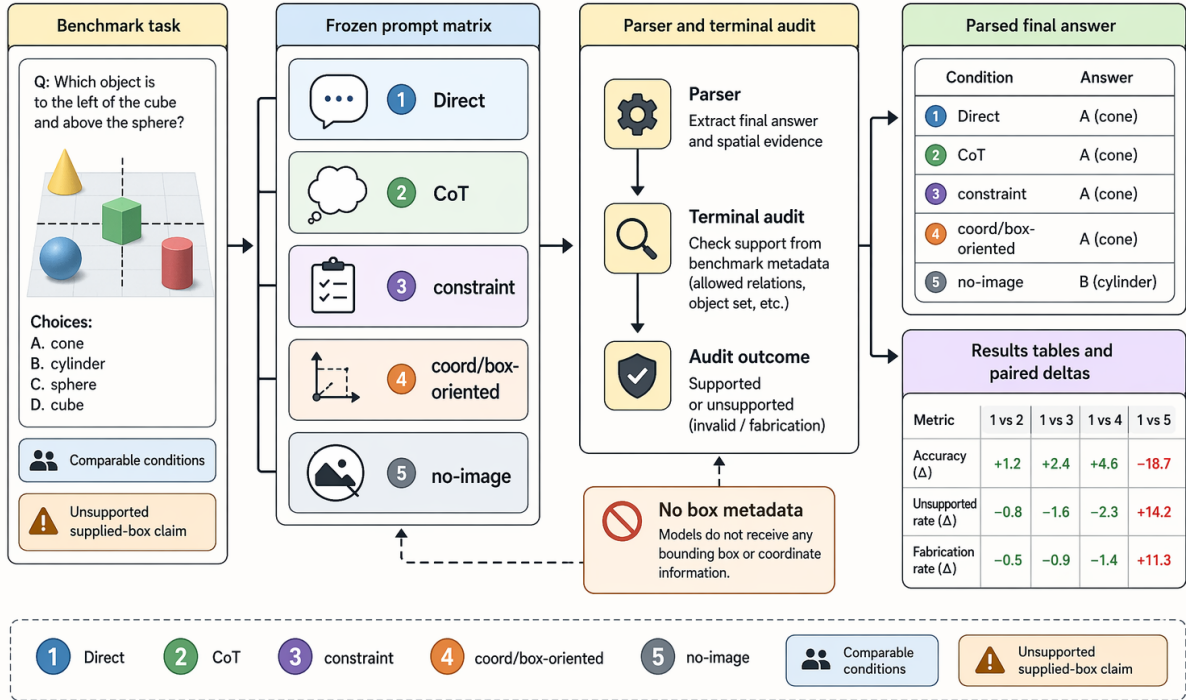


Figure 1: Train-free diagnostic pipeline. The same selected tasks are evaluated under five frozen inference-time conditions, parsed into a final answer, and scored by the benchmark rule. The coordinate/box-oriented branch is marked as no-supplied-box because the selected public releases provide zero box or coordinate rows.

task sets have different aggregate scores. This is especially important for spatial benchmarks because relation families and source strata have very different difficulty profiles.

Third, the box-source check is a design constraint rather than an after-the-fact caveat. The selected task releases contribute zero rows with supplied boxes and zero rows with supplied coordinates. The coordinate/box prompt is therefore not allowed to receive hidden geometry. Its name describes the reasoning format requested from the model, not an additional supervision channel. This is the main reason the paper reports a no-supplied-box diagnostic instead of a supplied-box grounding method.

Fourth, parse failures are handled conservatively. A response that cannot be normalized into the benchmark answer vocabulary remains incorrect. The four sentinel rows are reported because they affect exact-match accounting, but the paper does not remove them from headline accuracy or replace them with inferred answers. This convention protects the negative result from both directions: it avoids giving the coordinate/box prompt extra credit for malformed output, and it avoids blaming the entire null finding on a negligible parsing artifact.

Finally, the no-image prior is interpreted only as a lower-information diagnostic. It removes image access while leaving the question form and answer parser in place. The re-

sulting 34.98% pooled accuracy helps quantify text-only answerability, but it does not identify which visual examples contradict language priors. That stronger analysis would need audited counterfactual or minimal-pair grouping, which the current evidence pack does not contain.

Results

Table 1 gives the main result as a self-contained cross-benchmark matrix. The strongest visual condition is constraint-aware prompting at 78.70% pooled accuracy. Direct VLM scores 77.02%, natural-language CoT scores 77.80%, and the no-supplied-box coordinate/box-oriented prompt scores 76.91%. The no-image prior diagnostic is much lower at 34.98%, showing that image access matters for the selected tasks, but this alone does not establish a verified prior-conflict subset.

The benchmark rows explain why the pooled null result should not be overread as a universal spatial-reasoning conclusion. On What’sUp, all four visual conditions score 99.51%, so that slice contributes a ceiling-effect check rather than separating prompting strategies. On SpatialMQA, constraint-aware prompting is highest at 66.25%, while direct, natural-language CoT, and no-supplied-box coordinate/box prompting are all 62.08%. On SRBench, the best row is natural-language CoT at 56.25%, with the no-supplied-box

Source	Tasks	Backend	Metric	Budget	Accuracies by condition	Fails
Pooled	892	gpt-5-mini VLM	Exact match	512 tok., low effort	Direct 77.02; CoT 77.80; Constraint 78.70 ; Coord/box 76.91; No-image 34.98	1/0/2/1/0
SpatialMQA test	240	gpt-5-mini VLM	Exact match	512 tok., low effort	Direct 62.08; CoT 62.08; Constraint 66.25 ; Coord/box 62.08; No-image 30.42	0/0/1/0/0
SRBench test	240	gpt-5-mini VLM	Exact match	512 tok., low effort	Direct 53.33; CoT 56.25 ; Constraint 55.42; Coord/box 52.92; No-image 27.50	1/0/1/1/0
What'sUp ctrl.	412	gpt-5-mini VLM	First-caption	512 tok., low effort	Direct 99.51 ; CoT 99.51 ; Constraint 99.51 ; Coord/box 99.51 ; No-image 41.99	0/0/0/0/0

Table 1: The 76.91% no-supplied-box coordinate/box prompt does not beat the 77.02% direct baseline or the 78.70% constraint-aware condition. The backend is the gpt-5-mini hosted VLM for all rows; values are accuracies in percent and failure counts are ordered Direct/CoT/Constraint/Coord-box/No-image.

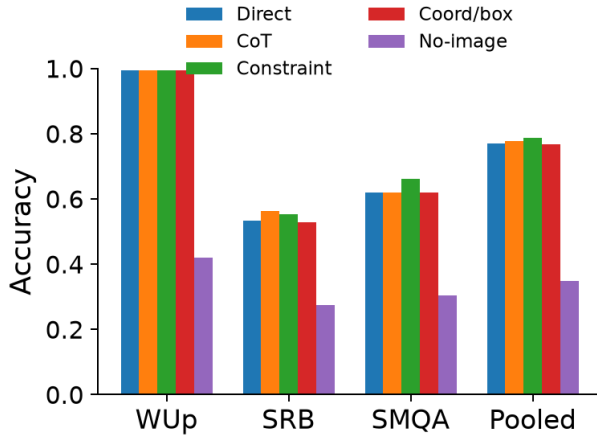


Figure 2: Family-balanced accuracy comparison. What'sUp is near ceiling for visual methods, while SRBench and SpatialMQA drive most differences between conditions.

coordinate/box condition at 52.92%. The aggregate therefore combines a near-ceiling controlled-image slice with two harder spatial-question slices where the coordinate/box-oriented prompt still does not produce a reliable advantage.

The paired tests are also null for the coordinate/box-oriented condition. Against direct VLM, its pooled delta is -0.11 percentage points with a 95% bootstrap interval of [-1.79, 1.57] and Holm-adjusted McNemar $p = 1.000000$. Against the strongest visual baseline, constraint-aware prompting, its pooled delta is -1.79 percentage points with interval [-3.59, 0.00] and Holm-adjusted $p = 1.000000$. This is not evidence for a supplied-box grounding improvement; it is evidence that the no-supplied-box prompt did not help in the current matrix.

The paired table and Figure 3 separate two claims that are easy to conflate. First, visual access is useful: direct VLM exceeds the no-image prior by 42.04 points, and the coordinate/box prompt exceeds no-image by 41.93 points.

Comparison	Delta pp	95% CI pp
Coord/box vs Direct	-0.11	[-1.79, 1.57]
Coord/box vs CoT	-0.90	[-2.69, 0.90]
Coord/box vs Constraint	-1.79	[-3.59, 0.00]
Coord/box vs No-image	41.93	[37.89, 45.96]
Direct vs No-image	42.04	[38.12, 46.08]

Table 2: Pooled paired deltas show a 41.93-point coord/box advantage over no-image but no improvement over visual baselines. Positive values favor the first condition; all visual-baseline comparisons have Holm-adjusted $p = 1.000000$, while no-image comparisons have Holm-adjusted $p = 0.000000$.

Second, the coordinate/box-oriented prompt does not add measurable value over ordinary visual prompting in this no-box setting. The visual-versus-no-image gap is therefore a task-evidence result, not proof that the coordinate/box prompt induced object localization.

Relation and Failure Analysis

The family-level table hides large relation and source-stratum differences. In SpatialMQA, constraint-aware prompting is strongest on behind, in-front-of, on/above, and right-of groups, while the no-supplied-box prompt is strongest only on left-of in the measured relation table. In SRBench, all visual methods struggle on the MRT easy and MRT hard strata, while navigation and orientation are much easier. In What'sUp, all visual methods are near ceiling across left-of, right-of, on, and under, but the no-image prior varies substantially by relation. Appendix Table 4 preserves representative slice values without crowding the main body.

The descriptive slices identify where future pair-audited work should look first. SpatialMQA left-of is the clearest local exception for the coordinate/box-oriented prompt, at 62.71% versus 50.85% for direct VLM, but the same prompt is lower than direct VLM on behind and right-of. SRBench exposes a different pattern: navigation is near solved for visual methods at 97.50%, while MRT hard remains close to

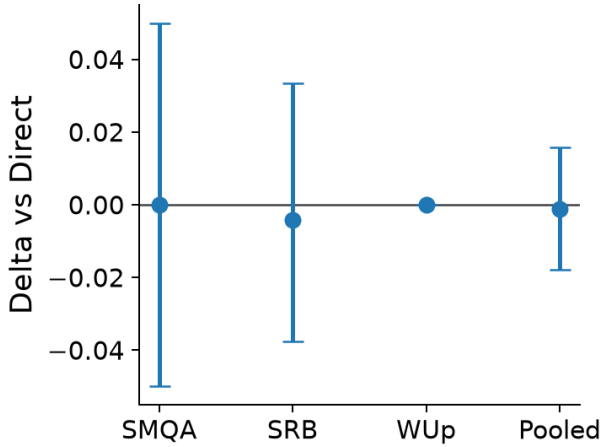


Figure 3: Paired deltas against direct VLM. The coordinate/box-oriented prompt’s pooled interval crosses zero, while visual conditions are consistently above the no-image prior diagnostic.

floor for direct VLM and coordinate/box prompting. These contrasts argue for relation- and source-aware diagnostics, not for a single prompt-level mechanism claim.

These breakdowns are descriptive. The scored examples contain relation-family tags, but they do not contain a verified flip-pair or minimal-pair grouping used for consistency metrics. Consequently the manuscript does not claim flip consistency, prior-conflict accuracy, or causal evidence of object-prior reliance. Those claims require verified pair identifiers and additional evaluation.

Parse failures are rare and do not explain the headline result. The evaluation records four unrecoverable parse failures: one for direct VLM on SRBench, one for constraint-aware prompting on SpatialMQA, one for constraint-aware prompting on SRBench, and one for the coordinate/box-oriented prompt on SRBench. All four are scored incorrect. Excluding them is a diagnostic sensitivity check only and does not replace the official accuracy; with only four sentinel rows among 4,460 responses, the negative/null coordinate result is not a parsing artifact. Appendix Figure 4 provides the compact method-level count plot.

Discussion

What the null result says. The central result is negative but still informative because the comparison is paired and complete. Every selected task has a scored response from direct VLM, natural-language CoT, constraint-aware prompting, no-supplied-box coordinate/box prompting, and the no-image prior. Under that design, the coordinate/box-oriented prompt is not penalized by a different task mix or by missing rows. It simply does not improve the pooled score: it is one task-level error behind direct VLM and 16 task-level errors behind constraint-aware prompting. The corresponding paired intervals include zero for visual-baseline comparisons, so the paper treats the result as a measured null rather than as a

small hidden gain.

This framing matters for spatial-reasoning papers because prompt names can imply mechanisms that the evidence does not isolate. A prompt that asks a model to think in coordinates is not the same thing as a system that receives object boxes, verifies those boxes, and computes relation labels from them. The selected public releases provide no supplied coordinate or box rows, and this run adds no detector. The coordinate/box condition is therefore a prompt-only intervention. Its failure to improve over direct visual answering says that this prompt shape was not sufficient in the measured setting; it does not rule out an actual supplied-box pipeline.

Why image access still matters. The no-image prior diagnostic prevents the null result from collapsing into the claim that these tasks are text-only. Across the pooled matrix, direct VLM is 42.04 points above no-image, and the coordinate/box prompt is 41.93 points above no-image. The visual signal is especially clear on What’sUp, where all visual methods reach 99.51% while the no-image prior reaches 41.99%. SR-Bench and SpatialMQA are harder, but both still show large visual-versus-no-image gaps at the family level.

At the same time, the no-image row is not a prior-conflict metric. It measures answerability when the image is removed, not whether the model ignored a contradictory image when one was present. Establishing prior conflict would require verified subsets where the textual prior and visual relation disagree, plus paired identifiers connecting counterfactual images or captions. The current data do not include that audit, so the paper uses the no-image condition only to bound text-only answerability.

Where the benchmark mixture is informative. The three benchmark families play different roles in the evidence. What’sUp confirms that the selected controlled images can be answered by the visual backend at near-ceiling accuracy, but ceiling behavior makes it a weak separator of prompt strategies. SpatialMQA is the slice where constraint-aware prompting gives the clearest family-level advantage, reaching 66.25% compared with 62.08% for direct VLM and the no-supplied-box coordinate/box prompt. SRBench is the slice where natural-language CoT is highest at 56.25%, but all visual methods remain far below their What’sUp accuracy.

The relation table adds a second caution. In SpatialMQA left-of, the coordinate/box prompt is higher than direct VLM by 11.86 points, but it is lower on behind and right-of. In SRBench, navigation is near solved for direct and coordinate/box prompting, while MRT hard remains near floor and the no-image prior is not uniformly lower than visual methods. These local patterns are useful for designing the next experiment, but they are not stable enough to support a broad mechanism claim from the present matrix.

Failure accounting. The parse-failure audit closes a mundane alternative explanation. Four of 4,460 responses were unrecoverable parse failures, and all were retained as incorrect. The failures touch direct VLM, constraint-aware prompting, and the coordinate/box prompt, but not enough rows to move the headline comparison. This is why the paper reports parse diagnostics separately from accuracy: they doc-

ument score semantics without turning parser cleanup into a post-hoc performance adjustment.

What evidence would change the claim. A stronger grounding paper would need a different evidence pack. First, a supplied-box or detected-box condition must actually receive boxes or coordinates, and the paper must document their source and coverage. Second, relation-consistency claims need audited minimal-pair or flip-pair group identifiers, not only relation-family labels. Third, mechanism claims need component ablations, such as centers-only, boxes-without-image, detected boxes, and supplied boxes. Fourth, robustness claims need repeated runs or backend comparisons. Without these additions, the honest contribution is the diagnostic boundary: the current no-supplied-box prompt does not improve spatial accuracy over visual baselines, while image access itself remains important.

Evaluation implications. The result suggests a practical review standard for spatial-reasoning claims. A paper that names boxes, coordinates, or grounding should show the channel that supplied those quantities and report coverage over the evaluated examples. A paper that claims consistency should report the grouping rule that defines relation flips or minimal pairs. A paper that claims robustness should separate paired task uncertainty from route, seed, or backend variation. The current manuscript follows those rules in the negative direction: it reports what was measured, marks unsupported claim families as missing or weak, and leaves stronger causal language out of the main result.

The same standard also helps interpret prompt engineering baselines. Natural-language CoT and constraint-aware prompting can be useful comparison conditions, but neither automatically explains why an answer changed. In this run, constraint-aware prompting is the best pooled visual condition, yet the evidence does not isolate whether the gain comes from relation checking, answer-format discipline, or benchmark-specific response behavior. The paper therefore reports it as a prompt baseline and keeps mechanism language for future ablation work.

Why the negative result is still a contribution. Negative spatial-reasoning results are easy to discard when they do not produce a new method winner. Here the null result is useful because it closes a specific tempting interpretation. A coordinate/box-oriented prompt sounds like a lightweight route to grounding, but when no boxes are actually supplied, it does not outperform direct visual answering on the paired matrix. That finding redirects future work toward evidence-bearing variants: official-box slices, detector-box pipelines, and pair-audited relation tests. It also prevents the paper from converting a prompt label into an unsupported localization claim.

Benchmark-design lessons. The family differences also show why a single spatial benchmark is not enough for this question. What’sUp is valuable because it controls object identity while varying spatial relations, but the visual backend is already near ceiling on the selected controlled images. SR-Bench and SpatialMQA are harder and expose more variation among prompts, yet their source strata and relation labels are

not the same object-level pairing signal as a verified counterfactual set. A diagnostic paper therefore needs both breadth and auditability: breadth to avoid drawing conclusions from one benchmark’s ceiling or floor, and auditability to connect a score difference to a specific spatial contrast. This benchmark-design caution follows directly from prior spatial evaluation resources: controlled relation variation, broader MLLM spatial question sets, and source-stratified spatial probes each expose different failure surfaces rather than a single universal spatial score (Kamath, Hessel, and Chang 2023; Liu et al. 2025; Stogiannidis, McDonagh, and Tsaftaris 2025).

This paper has the breadth side but only partial auditability. It covers three public benchmark families and five paired inference conditions, but it does not have box metadata, detector outputs, or verified flip groups. The appropriate conclusion is correspondingly narrow. The current matrix can say that no-supplied-box coordinate prompting failed to beat visual baselines, that image access improved over text-only answering, and that relation/source slices vary substantially. It cannot say that the model succeeded or failed at explicit object grounding.

Conclusion

This paper reports an evidence-bounded train-free diagnostic for spatial-relation VLM behavior. On the selected public releases, image access matters, but a no-supplied-box coordinate/box-oriented prompt does not improve over direct visual answering and trails constraint-aware prompting. The result is a negative/null finding for the planned supplied-box story and a concrete specification of the missing evidence: future work needs official or detected boxes, verified flip/minimal-pair groups, and component ablations before making stronger grounding claims.

Limitations

The main limitation is box-source availability. The selected releases provide no supplied boxes, coordinates, detector boxes, verified flip-pair identifiers, component ablations, or repeated backend runs. The coordinate/box condition is therefore a prompt-shape diagnostic, not a localization-grounded method, and the paper avoids supplied-box, prior-conflict, flip-consistency, ablation, and robustness claims.

The pooled metric also inherits benchmark heterogeneity. What’sUp, SRBench, and SpatialMQA differ in construction, answer spaces, relation labels, and difficulty, so the pooled score is useful only when read with the family and relation slices (Kamath, Hessel, and Chang 2023; Stogiannidis, McDonagh, and Tsaftaris 2025; Liu et al. 2025). This work evaluates public benchmark tasks and model outputs for measurement, not deployment-time decision support.

These limits shape the intended use of the artifact. The benchmark rows, prompts, parser rules, and aggregate tables support audit and reproduction of the reported null result, but they should not be used as a deployed spatial-reasoning system or as evidence about individual images outside the selected public tasks.

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Reproducibility Checklist

The analysis uses scored model responses, a frozen five-condition method matrix, benchmark-family provenance records, box-availability records, result tables, paired-delta tables, relation-breakdown tables, and parse-failure diagnostics. These artifact classes constrain the paper claims: scored responses establish completion, box-availability records enforce the no-supplied-box limitation, result and paired-delta tables support the accuracy and interval claims, relation breakdowns support descriptive slice analysis, and parse diagnostics document sentinel rows. Additional supplied-box, detected-box, flip-consistency, ablation, or robustness claims should remain out of scope until new evidence is produced and checked. Reproduction should respect the licenses and redistribution rules of the public benchmark sources, and any release of derived examples should separate benchmark content from model responses. Raw operational traces, private service details, and local execution configuration are not scientific evidence and should remain outside the manuscript.

Claim family	Current evidence	Paper action
Supplied boxes	0 selected box/coordinate rows	Report no-supplied-box prompt only
Flip consistency	No verified pair ids	Treat relation slices as descriptive
Component ablations	No box, detector, or center variants	Avoid mechanism attribution
Robustness	One completed evaluation pass	Avoid seed or backend robustness claims

Table 3: Four missing evidence families bound the claim scope. The appendix ledger records why the main text reports a negative/null no-supplied-box diagnostic rather than supplied-box grounding, flip consistency, ablation, or robustness claims.

Parse Diagnostics

Parse diagnostics document score accounting rather than define a separate result. The plot below shows the sentinel rows retained as incorrect, confirming that malformed outputs are rare and distributed across methods rather than concentrated in the coordinate/box condition.

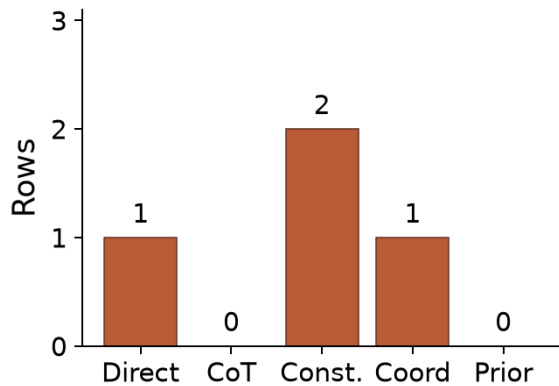


Figure 4: Parse failures by method. Four of 4,460 rows were unrecoverable parse failures and are retained as incorrect in the reported accuracies.

Representative Relation Slices

The slice table records representative relation-family and source-stratum rows used in the discussion. These rows are descriptive: they show heterogeneity across families and relations, but they are not audited flip pairs and do not support a consistency metric.

Family	Group	Direct	Coord/box
SpatialMQA	behind	60.53	57.89
SpatialMQA	left-of	50.85	62.71
SpatialMQA	right-of	61.43	55.71
SRBench	MRT hard	2.50	5.00
SRBench	navigation	97.50	97.50
SRBench	orientation	80.00	77.50
What'sUp	left-of	99.03	99.03
What'sUp	under	100.00	100.00

Table 4: Representative relation-family and source-stratum accuracies. The table is descriptive and does not define a flip-consistency metric.